Kennesaw State University
Department of Electrical and Computer Engineering

EE 6530 Antenna
Chapter 2 Official Presentation Transcript

Antenna Fundamentals

Prepared by: Hossein Molhem

Textbook: Stutzman and Thiele, *Antenna Theory and Design*, 3rd Edition

Contents

EE 6530 Antenna — Chapter 2 Official Presentation Transcript

Chapter 2: Antenna Fundamentals

Textbook: Stutzman and Thiele, *Antenna Theory and Design*, 3rd Edition
Lecture Theme: Fields, radiation patterns, directivity, gain, impedance, efficiency, and polarization

Slide 1: Chapter 2 — Antenna Fundamentals

Today we begin Chapter 2, *Antenna Fundamentals*. In Chapter 1, we developed the broad physical picture of antennas: an antenna is the interface between a guided electromagnetic wave and a free-space electromagnetic wave. We also discussed how time-varying charge and current distributions produce radiation.

In Chapter 2, we move from physical description to mathematical and engineering analysis. The goal is to understand how current on an antenna creates electric and magnetic fields, how those fields behave in the far-field region, and how we define the main antenna parameters: radiation pattern, directivity, gain, input impedance, radiation efficiency, and polarization.

This chapter is fundamental for everything that follows. Dipoles, monopoles, arrays, aperture antennas, radar antennas, MIMO systems, and direction-of-arrival estimation all depend on the concepts introduced here.

Slide 2: Lecture Roadmap

This slide shows the roadmap for the lecture. The logic of the chapter is very important.

We begin with Maxwell's equations in phasor form. Then we introduce the vector potential formulation, which gives us a practical way to compute radiation from a known current distribution. After that, we study the ideal dipole, which is the simplest analytical radiator in antenna theory.

Once the ideal dipole is understood, we introduce the main engineering parameters: radiation pattern, half-power beamwidth, directivity, gain, input impedance, radiation resistance, efficiency, and polarization.

The central question of the chapter is this:

If the current distribution on an antenna is known, how do we predict the radiated fields and the antenna parameters?

The analytical chain is:

$$J \to A \to E, H \to F(\theta, \phi), D, G, Z_A, e_r, \text{polarization.}$$

Slide 3: Why Chapter 2 Matters

Chapter 2 matters because it connects two viewpoints of the same physical device.

From the electromagnetic viewpoint, an antenna is a current distribution that produces electric and magnetic fields. From the circuit viewpoint, the same antenna is a load with an input impedance. These two viewpoints are not separate; they are connected through power flow, radiation resistance, and efficiency.

In practical RF engineering, it is not enough to know only the gain of an antenna. We must also know the pattern, impedance, efficiency, bandwidth, and polarization. For radar and array systems, we must also understand how the antenna modifies the amplitude and phase of the received signal before the signal reaches the receiver and the ADC.

So the purpose of this chapter is to build the language needed to describe an antenna both as an electromagnetic radiator and as a component in an RF system.

Slide 4: Time-Harmonic Field Assumption

Most antenna analysis assumes sinusoidal steady-state excitation. This means the fields vary harmonically in time at angular frequency ω.

Instead of writing the full time-domain electric field, we use phasor notation:

$$\mathcal{E}(\mathbf{r}, t) = \Re\{E(\mathbf{r})e^{j\omega t}\}.$$

This is the same basic idea used in AC circuit analysis. The time dependence is represented by $e^{j\omega t}$, and the spatial dependence is contained in the complex phasor $E(\mathbf{r})$.

The key mathematical advantage is that differentiation with respect to time becomes multiplication by $j\omega$. This converts Maxwell's equations from time-domain differential equations into phasor-domain equations that are easier to solve for antennas.

Slide 5: Maxwell Equations in Phasor Form

This slide gives Maxwell's equations in phasor form. For antenna radiation problems, the two curl equations are especially important:

$$\nabla \times E = -j\omega\mu H,$$

$$\nabla \times H = j\omega\epsilon E + J.$$

The current density J is the impressed source current. In antenna theory, this is usually the current on the wire, surface, or aperture that produces radiation.

The divergence equations describe the source charge and the absence of magnetic charge. The continuity equation,

$$\nabla \cdot J = -j\omega\rho,$$

ensures conservation of charge.

The important engineering message is that radiation is caused by time-varying current and charge. If we can model the current distribution on the antenna, we can calculate the fields.

Slide 6: Power Flow — The Poynting Vector

The Poynting vector connects fields to power:

$$S = \frac{1}{2}E \times H^*.$$

This is a complex power density. Its real part represents time-average real power flow in watts per square meter.

For antennas, radiated power is calculated by integrating the outward power density over a closed surface surrounding the antenna:

$$P = \frac{1}{2}\Re\left\{\iint_s (E \times H^*) \cdot ds\right\}.$$

In the far field, the electric and magnetic fields decay as $1/r$. Therefore, the power density decays as $1/r^2$. But the area of a sphere increases as $4\pi r^2$. These two effects cancel, so the total radiated power crossing any far-field sphere remains constant.

This is why radiated power is conserved as the wave propagates outward.

Slide 7: Vector Potential Formulation

Solving Maxwell's equations directly is usually difficult. The vector potential A gives us a practical method.

We define

$$H = \frac{1}{\mu}\nabla \times A.$$

Then the vector potential satisfies a wave equation driven by the source current density:

$$\nabla^2 A + \beta^2 A = -\mu J.$$

For a general current distribution, the solution is

$$A = \iiint_{v'} \mu J(\mathbf{r}')\frac{e^{-j\beta R}}{4\pi R} dv'.$$

This equation has a clear physical interpretation. Every small piece of antenna current contributes a small spherical wave to the observation point. The term $1/R$ represents amplitude spreading. The term $e^{-j\beta R}$ represents phase delay due to propagation.

This phase term is extremely important in arrays, beamforming, MIMO radar, and DOA estimation.

Slide 8: Far-Field Approximation

In the far field, the observation point is much farther away than the physical size of the antenna. Under this condition, the distance from each current element to the observation point can be approximated as

$$R \approx r - \hat{r} \cdot \mathbf{r}'.$$

This approximation separates the common spherical spreading term from the angular phase term. The far-field vector potential becomes

$$A = \mu\frac{e^{-j\beta r}}{4\pi r} \iiint_{v'} J(\mathbf{r}')e^{j\beta\hat{r}\cdot\mathbf{r}'} dv'.$$

The far field has local plane-wave behavior:

$$E \perp H, \qquad E \perp \hat{r}, \qquad H \perp \hat{r},$$

and

$$\frac{|E|}{|H|} = \eta.$$

So even though the wave spreads spherically, a small region of the wavefront looks locally like a plane wave.

Slide 9: The Ideal Dipole

The ideal dipole is the simplest analytical antenna. It is a very short current element of length Δz, where

$$\Delta z \ll \lambda.$$

The current is assumed uniform in magnitude and phase over the whole element:

$$I(z) = I.$$

This is not a perfectly realistic antenna, because real wire currents usually taper to zero at the ends. However, the ideal dipole is extremely useful because it provides a closed-form solution and reveals the basic physics of radiation.

Any practical current distribution can be conceptually divided into many small current elements. The total field is then found by summing the contribution from all of them. This idea is the basis of many analytical and numerical antenna methods.

Slide 10: Far Fields of the Ideal Dipole

The far fields of a z-directed ideal dipole are

$$E_\theta = j\omega\mu \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin\theta,$$

$$H_\phi = j\beta \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin\theta.$$

There are several important features here.

First, both fields decay as $1/r$, which is the signature of radiation fields.

Second, both fields contain the propagation phase factor $e^{-j\beta r}$.

Third, the angular dependence is $\sin\theta$. Therefore, the radiation is zero along the dipole axis, at $\theta = 0^\circ$ and 180°, and maximum broadside to the dipole, at $\theta = 90^\circ$.

Finally,

$$\frac{E_\theta}{H_\phi} = \eta.$$

This confirms that the far field behaves locally as a transverse electromagnetic wave.

Slide 11: Numerical Example 1 — Ideal Dipole Fields

In this numerical example, we make the ideal dipole fields concrete.

At

$$f = 300 \text{ MHz},$$

the wavelength is

$$\lambda = 1 \text{ m}.$$

The dipole length is

$$\Delta z = \frac{\lambda}{50} = 0.02 \text{ m}.$$

For a peak current of 1 A, at $r = 100$ m and $\theta = 90^\circ$, the electric field magnitude is

$$|E_\theta| = 0.0377 \text{ V/m}.$$

The magnetic field is

$$|H_\phi| = 1.0 \times 10^{-4} \text{ A/m}.$$

The radiation resistance is

$$R_r = 0.316 \ \Omega.$$

The most important lesson is that a very short dipole has very small radiation resistance. That makes matching and efficiency difficult in practical electrically small antennas.

Slide 12: Field Regions Around an Antenna

The electromagnetic field around an antenna is divided into regions.

The first region is the reactive near field. In this region, energy is mainly stored near the antenna and exchanged between electric and magnetic fields. This is analogous to reactive energy in capacitors and inductors.

The second region is the radiating near field. In this region, radiated power exists, but the angular pattern may still depend on distance.

The third region is the far field. In the far field, the radiation pattern depends only on direction, not on distance. The fields are transverse and behave locally as plane waves.

For electrically small antennas, the reactive near field extends approximately to

$$r \approx \frac{\lambda}{2\pi}.$$

For larger antennas, a commonly used far-field criterion is

$$r_{ff} \approx \frac{2D^2}{\lambda},$$

where D is the largest antenna dimension.

Slide 13: Radiation Pattern Definitions

The radiation pattern describes how the far-field radiation varies with angle.

The normalized field pattern is

$$F(\theta, \phi) = \frac{E(\theta, \phi)}{E_{\max}}.$$

The normalized power pattern is

$$P(\theta, \phi) = |F(\theta, \phi)|^2.$$

The half-power points are the angles where the power drops to one half of the maximum value. Since power is proportional to the square of field magnitude, the field magnitude at the half-power point is

$$|F| = \frac{1}{\sqrt{2}} \approx 0.707.$$

In decibels, this is

$$-3 \text{ dB}.$$

This is why half-power beamwidth is also called the 3-dB beamwidth.

Slide 14: Numerical Example 2 — Pattern and HPBW

For the ideal dipole,

$$F(\theta) = \sin\theta.$$

The normalized power pattern is

$$P(\theta) = \sin^2\theta.$$

The half-power condition is

$$\sin^2\theta = \frac{1}{2}.$$

This occurs at

$$\theta = 45^\circ$$

and

$$\theta = 135^\circ.$$

Therefore, the half-power beamwidth is

$$\text{HPBW} = 135^\circ - 45^\circ = 90^\circ.$$

This confirms the broad doughnut-shaped radiation pattern of the ideal dipole.

Slide 15: Radiation from Line Currents

Many antennas can be approximated as line currents. For a z-directed line current, the far-field vector potential contains the integral

$$A_z = \mu \frac{e^{-j\beta r}}{4\pi r} \int I(z') e^{j\beta z' \cos\theta} dz'.$$

This equation is extremely important. It tells us that the far field is the coherent sum of contributions from all points along the current distribution.

The exponential term represents the phase difference between different source points. These phase differences create constructive interference in some directions and destructive interference in others. That is how main lobes, side lobes, and nulls are formed.

This is also the mathematical foundation of array beamforming.

Slide 16: Element Factor and Pattern Factor

A radiation pattern can often be written as

$$F(\theta, \phi) = g(\theta, \phi) f(\theta, \phi).$$

Here g is the element factor, and f is the pattern factor.

The element factor describes how a single small radiator behaves. For a z-directed current element,

$$g(\theta) = \sin\theta.$$

The pattern factor describes how the spatial distribution of current produces interference.

In antenna arrays, this becomes the pattern multiplication principle:

$$F_{\text{total}}(\theta, \phi) = F_{\text{element}}(\theta, \phi) F_{\text{array}}(\theta, \phi).$$

This is very important in MIMO radar and DOA estimation because the array response is not only a function of element spacing. It is also shaped by the individual element pattern.

Slide 17: Directivity

Directivity measures how strongly an antenna concentrates radiation in a particular direction compared with an isotropic radiator.

The radiation intensity is

$$U(\theta,\phi) = r^2 S(\theta,\phi).$$

The average radiation intensity is

$$U_{\text{ave}} = \frac{P_{\text{rad}}}{4\pi}.$$

The maximum directivity is

$$D = \frac{U_{\text{max}}}{U_{\text{ave}}}.$$

Using the beam solid angle Ω_A, directivity can also be written as

$$D = \frac{4\pi}{\Omega_A}.$$

This equation gives the main physical interpretation: a narrower beam has a smaller beam solid angle and therefore a larger directivity.

Directivity is determined only by pattern shape. It does not include loss.

Slide 18: Numerical Example 3 — Directivity and Gain

For the ideal dipole,

$$F(\theta) = \sin\theta.$$

The beam solid angle is

$$\Omega_A = \frac{8\pi}{3}.$$

Therefore,

$$D = \frac{4\pi}{8\pi/3} = \frac{3}{2} = 1.5.$$

In decibels,

$$D = 1.76 \text{ dBi}.$$

If the radiation efficiency is

$$e_r = 0.85,$$

then the gain is

$$G = e_r D = 0.85(1.5) = 1.275.$$

In dBi,

$$G = 1.06 \text{ dBi}.$$

This example shows that gain is directivity reduced by losses.

Slide 19: Gain versus Directivity

This slide emphasizes a common distinction.

Directivity is a purely geometrical or pattern-based quantity. It asks: how concentrated is the radiated power in a certain direction?

Gain includes radiation efficiency. It asks: how much of the accepted input power becomes useful radiation in the desired direction?

The relation is

$$G = e_r D.$$

If the antenna is lossless, then $e_r = 1$, and $G = D$. But for a real lossy antenna, $G < D$.

In radar and communication system calculations, gain is often more practical because it accounts for both focusing and loss.

Slide 20: Antenna Input Impedance

From the circuit viewpoint, the antenna is represented by an input impedance:

$$Z_A = R_A + jX_A.$$

The real part is

$$R_A = R_r + R_o.$$

Here R_r is radiation resistance, representing power that leaves the antenna as radiation. R_o is ohmic loss resistance, representing power converted into heat.

The imaginary part X_A represents reactive energy stored near the antenna.

This impedance is crucial because the antenna must be matched to the transmission line or source. In many RF systems, the reference impedance is

$$Z_0 = 50\ \Omega.$$

If Z_A is very different from 50 Ω, a large fraction of the incident power is reflected.

Slide 21: Radiation Resistance

Radiation resistance is one of the most important concepts in antenna theory.

It allows us to represent radiated power using a circuit model:

$$P_{\text{rad}} = \frac{1}{2} R_r |I_A|^2.$$

For an ideal dipole,

$$R_r = 80\pi^2 \left(\frac{\Delta z}{\lambda} \right)^2 \ \Omega.$$

This shows that radiation resistance scales with the square of electrical length.

If $\Delta z \ll \lambda$, then R_r becomes very small. This is why electrically small antennas are difficult: they may have low radiation resistance and large reactance, making both efficiency and impedance matching challenging.

Slide 22: Numerical Example 4 — Short Dipole Impedance

In this example, a short dipole operates at

$$f = 30 \text{ MHz}.$$

The wavelength is

$$\lambda = 10 \text{ m}.$$

For a total length of 0.5 m, the electrical length is

$$\frac{L}{\lambda} = 0.05.$$

For a practical short dipole,

$$R_r = 20\pi^2 \left(\frac{L}{\lambda} \right)^2 = 0.494 \ \Omega.$$

With ohmic loss resistance

$$R_o = 0.20 \ \Omega,$$

the radiation efficiency is

$$e_r = \frac{R_r}{R_r + R_o} = 0.712.$$

The assumed input impedance is

$$Z_A = 0.694 - j600\ \Omega.$$

So the antenna may have reasonable radiation efficiency after accepting power, but direct connection to a 50 Ω line would be a severe mismatch.

Slide 23: Efficiency and Matching Are Different

This slide corrects a frequent misunderstanding.

Radiation efficiency tells us what fraction of accepted power is radiated:

$$e_r = \frac{P_{\text{rad}}}{P_{\text{in}}}.$$

Impedance matching tells us how much incident power is accepted by the antenna in the first place.

An antenna can have acceptable radiation efficiency but still be badly mismatched to a 50 Ω system. In that case, most incident power is reflected before it is ever accepted by the antenna.

This is why small antennas usually require matching networks. The matching network must cancel the large reactance and transform the small resistance to the system impedance.

Slide 24: Polarization

Polarization describes the time-varying orientation of the electric field vector.

If the tip of the electric field vector moves along a fixed line, the wave is linearly polarized.

If the tip rotates in a circle, the wave is circularly polarized.

In the most general case, the tip traces an ellipse, and the wave is elliptically polarized.

Polarization is important because a receiving antenna responds only to the component of the electric field aligned with its own polarization. If the transmitting and receiving polarizations are mismatched, the received power is reduced.

For a vertical dipole, the radiated electric field is vertically polarized in the usual broadside direction.

Slide 25: Numerical Example 5 — Polarization Mismatch

Consider a vertically polarized transmitting antenna and a receiving antenna tilted by

$$\psi = 45^\circ.$$

For two linearly polarized antennas, the polarization loss factor is

$$\text{PLF} = \cos^2 \psi.$$

Therefore,

$$\text{PLF} = \cos^2(45^\circ) = 0.5.$$

In decibels,

$$10 \log_{10}(0.5) = -3.01 \text{ dB}.$$

So a 45° polarization mismatch causes a 3-dB power loss. That means half of the available received power is lost because of polarization misalignment alone.

Slide 26: Linear, Circular, and Elliptical Cases

For a wave propagating in the $+z$-direction, the transverse electric field can be written as

$$E = E_x \hat{x} + E_y \hat{y}.$$

If only one component exists, or if the two components remain in phase or 180 degrees out of phase, the polarization is linear.

If the two orthogonal components have equal amplitudes and a 90° phase difference, the polarization is circular.

If the amplitudes are unequal, or if the phase difference is not one of the special linear or circular cases, the polarization is elliptical.

In radar, polarization is not only a loss mechanism. It can also carry information about target shape, orientation, and scattering behavior.

Slide 27: Connection to Radar Systems

Now we connect Chapter 2 to radar.

In radar, antenna gain determines how much transmitted power is concentrated toward the target and how much scattered power is collected on receive. Radiation pattern determines angular coverage, clutter illumination, and spatial selectivity.

Polarization affects target scattering. Some targets respond differently to vertical, horizontal, circular, or dual-polarized waves.

Input impedance and efficiency affect how much transmitter power is actually radiated.

Therefore, the radar equation is not just a signal-processing equation. It depends strongly on antenna parameters.

Slide 28: Connection to Arrays, MIMO, and DOA

This slide is especially important for array processing.

In a simplified array model, we often write the steering vector using only phase differences caused by element spacing. But the real array response also includes the element pattern, polarization, mutual coupling, impedance effects, and calibration errors.

The total pattern is approximately

$$F_{\text{total}}(\theta, \phi) = F_{\text{element}}(\theta, \phi) F_{\text{array}}(\theta, \phi).$$

If the element pattern or coupling is ignored, the array manifold becomes distorted. That distortion affects the covariance matrix and degrades MUSIC, ESPRIT, beamforming, and MIMO radar estimation.

Thus, antenna fundamentals directly affect DOA accuracy.

Slide 29: Connection to FPGA/DSP Implementation

For FPGA and DSP implementation, the antenna is the electromagnetic front end.

The digital algorithm begins after the analog RF chain and ADC. But the signal reaching the ADC has already been shaped by the antenna pattern, gain, polarization, impedance match, efficiency, and mutual coupling.

A digital processor can estimate, filter, and accelerate computations, but it cannot fully recover information that was never captured correctly by the antenna system.

This is why antenna theory matters even for hardware acceleration and signal-processing research. The array covariance matrix used in DOA estimation is ultimately produced by the physical antenna system.

Slide 30: Teaching Summary — What Students Must Remember

After Chapter 2, students should remember the following points.

First, current distribution is the source of radiation.

Second, the vector potential provides a practical bridge from current to fields.

Third, in the far field, the wave is transverse and behaves locally like a plane wave.

Fourth, the ideal dipole has the basic pattern

$$F(\theta) = \sin\theta.$$

Fifth, directivity describes angular concentration, while gain includes efficiency.

Sixth, input impedance includes radiation resistance, ohmic loss resistance, and reactance.

Finally, polarization mismatch can reduce received power even if the antenna gain and impedance match appear acceptable.

Slide 31: Final Chapter 2 Message

The final message of Chapter 2 is that antenna analysis connects four major ideas:

$$\text{current} \rightarrow \text{fields} \rightarrow \text{patterns} \rightarrow \text{system performance}.$$

An antenna must be understood both electromagnetically and as an RF circuit element.

Electromagnetically, it converts current into radiated fields.

As a circuit element, it has impedance, accepts power, stores reactive energy, dissipates loss, and radiates useful power.

This chapter provides the foundation for practical dipoles, monopoles, arrays, aperture antennas, antenna measurements, radar systems, MIMO systems, and DOA estimation.

The main engineering lesson is this:

Antenna performance is not described by one number. It is described by the combined behavior of pattern, directivity, gain, impedance, efficiency, and polarization.

End of Transcript